Direction of vorticity and continuation for 3D Navier–Stokes: the $\frac{1}{2}$-Hölder criterion on the maximal classical interval

Benjamin Stanley Frohman*

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Abstract

Let u be the unique classical solution of the 3D incompressible Navier–Stokes equations on the maximal interval $[0, T^*)$, viscosity $\nu > 0$. Write $\omega = \nabla \times u$, $r = |\omega|$, $\xi = \omega/r$ on $\{r > 0\}$, and $S = \frac{1}{2}\big(\nabla u + (\nabla u)^\top\big)$. The two constants of the criterion are

$$\Omega = \Omega(u_0, \nu) > 0, \qquad C = C(u_0, \nu) < \infty.$$

If

$$|\xi(x,t) - \xi(y,t)| \le C\,|x-y|^{1/2}$$

whenever $r(x,t), r(y,t) \ge \Omega$ and $t < T^*$, then $T^* = \infty$ and u is globally smooth. The direction ξ is never evaluated on $\{r = 0\}$. No additional geometric term is added to the PDE. The implication is the Beirão da Veiga–Berselli $\frac{1}{2}$-Hölder form of the Constantin–Fefferman criterion. An attempt to construct Ω and C from (1)–(2) for arbitrary smooth divergence-free u_0 is carried out below. The resulting a priori bound on C does not close at the energy level. The test therefore does not decide global smoothness versus blow-up.

MSC-style keywords: Navier–Stokes, vorticity direction, Beale–Kato–Majda, Constantin–Fefferman, Beirão da Veiga–Berselli.

Contents

*No tether term is used. Degeneracy of $\xi = \omega/|\omega|$ is respected. Theorem A is the Beirão da Veiga–Berselli implication. Theorem B records the result of constructing Ω, C from the unmodified PDE: the a priori bound on C does not close, so neither global smoothness nor blow-up is obtained for arbitrary data.

1 Scope

This note treats the unmodified Cauchy problem for 3D incompressible Navier–Stokes, written as (1)–(2) below, on the maximal classical existence interval $[0, T^*)$. The only geometric input is a uniform spatial $\frac{1}{2}$-Hölder bound on the vorticity *direction* on the set where the vorticity *magnitude* is at least a positive threshold. From that input one concludes $T^* = \infty$.

The following are **not** proved: that $C(t)$ remains finite up to T^* for every smooth divergence-free u_0; that $T^* < \infty$ occurs for some such u_0; that a named tether or metriplectic correction is present in the equations. The PDE used is (1).

Section 9 constructs candidate constants from the PDE and tests whether the energy identity forces them to stay finite. It does not.

2 The equations

Let $\nu > 0$. On $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$ (the whole-space case is identical up to replacing the periodized Newtonian potential by $1/(4\pi|x|)$) one considers

$$\partial_t u + (u \cdot \nabla)u = \nu\Delta u - \nabla p, \tag{1}$$

$$\nabla \cdot u = 0, \tag{2}$$

with $u(\,\cdot\,, 0) = u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$, $\nabla \cdot u_0 = 0$. Local well-posedness in H^s, $s > 5/2$, yields a unique classical solution on a maximal interval $[0, T^*)$, $T^* \in (0, \infty]$, with the standard blow-up alternative: if $T^* < \infty$ then

$$\limsup_{t \uparrow T^*} \|\nabla u(t)\|_{L^2(\mathbb{T}^3)} = \infty. \tag{3}$$

Equivalently, $\limsup_{t\uparrow T^*} \|\omega(t)\|_{L^2} = \infty$, since $\|\omega\|_{L^2} = \|\nabla u\|_{L^2}$ under (2). This is the energy-class continuation criterion (Kato); it is the only continuation tool used below. (Beale–Kato–Majda in $L^1_t L^\infty_x$ is stronger than needed once enstrophy is bounded.)

Curl of (1), using (2), gives the vorticity equation

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu\Delta\omega, \qquad \omega = \nabla \times u. \tag{4}$$

Write $r := |\omega|$ and, on the open set $\{r > 0\}$,

$$\xi := \frac{\omega}{r}. \tag{5}$$

The strain is the symmetric gradient

$$S := \frac{1}{2}\big(\nabla u + (\nabla u)^\top\big). \tag{6}$$

On $\{r > 0\}$ one has the stretching identity

$$(\omega \cdot \nabla)u \cdot \omega = r^2\,(S\xi \cdot \xi). \tag{7}$$

The left-hand side of (7) extends continuously by 0 at $r = 0$. The unit field ξ is *not* extended; every estimate below that mentions $\xi(x)$ is restricted to $r(x) > 0$.

3 The two constants

The geometric criterion uses exactly two solution-class constants, both allowed to depend on the data (u_0, ν) and on nothing else from the later evolution:

$$\Omega = \Omega(u_0, \nu) > 0, \tag{8}$$

$$C = C(u_0, \nu) < \infty. \tag{9}$$

Here Ω is the vorticity-intensity threshold: ξ is constrained only on $\{r \geq \Omega\}$. The complementary set $\{r < \Omega\}$ is estimated without forming ξ. The constant C is the uniform spatial $\frac{1}{2}$-Hölder modulus of ξ on that high set. From (8)–(9) the enstrophy argument produces one derived majorant coefficient, depending on the two constants and on viscosity,

$$K(C, \Omega, \nu) = \frac{C_*}{\nu}\,(C + \Omega)^2, \tag{10}$$

where $C_* < \infty$ depends only on the Sobolev, Hardy–Littlewood–Sobolev, and Calderón–Zygmund constants of $\mathbb{T}^3$ (or $\mathbb{R}^3$). The pair (Ω, C) is the input; K is not an independent hypothesis.

These two constants are not constructed from (1) in this note. Theorem A takes (8)–(9) as given.

4 The criterion

Theorem A (Beirão da Veiga–Berselli, recast)**.** *Let u be the classical solution of* (1)–(2) *on* $[0, T^*)$*, and let $\Omega = \Omega(u_0, \nu)$ and $C = C(u_0, \nu)$ be the two constants* (8)–(9)*. Suppose that for all $t \in [0, T^*)$ and all $x, y \in \mathbb{T}^3$ with $r(x,t) \geq \Omega$ and $r(y,t) \geq \Omega$,*

$$|\xi(x,t) - \xi(y,t)| \leq C\,|x - y|^{1/2}. \tag{11}$$

Then $T^* = \infty$, and $u \in C^\infty(\mathbb{T}^3 \times [0, \infty))$. The continuation bound depends on the two constants only through $K(C, \Omega, \nu)$ in (10)*:*

$$\sup_{t<T^*} \|\omega(t)\|_{L^2}^2 \leq \|\omega(0)\|_{L^2}^2 \exp\Big(\frac{K(C, \Omega, \nu)}{2\nu}\,\|u_0\|_{L^2}^2\Big). \tag{12}$$

The Hölder condition may be localized to $|x - y| < \delta$ for an arbitrary $\delta > 0$; the far field is controlled by the integrable periodized kernel. Only pairs in which *both* magnitudes meet the threshold Ω are constrained. Pairs with $r(x) < \Omega$ or $r(y) < \Omega$ are estimated crudely, without using ξ.

Remark 1. On $\{r(x), r(y) > 0\}$ one has $|\xi(x) - \xi(y)| = 2|\sin(\theta/2)|$ and $\sin\theta \leq |\xi(x) - \xi(y)|$, where $\theta = \angle(\omega(x), \omega(y))$. Thus (11) implies $\sin\theta(x, y, t) \leq C|x - y|^{1/2}$ on the high-high set, which is Assumption A of Beirão da Veiga–Berselli [1] at exponent $\alpha = 1/2$ with g a constant. Lipschitz direction (Constantin–Fefferman [3]) is the case $\alpha = 1$.

5 Enstrophy and stretching

Take the L^2 inner product of (4) with ω. After an integration by parts (justified on every compact subinterval of $[0, T^*)$ by classical smoothness),

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu\|\nabla\omega\|_{L^2}^2 = \int_{\mathbb{T}^3} \mathcal{K}(x,t)\,\mathrm{d}x, \tag{13}$$

where the stretching density is

$$\mathcal{K}(x,t) := \big((\omega\cdot\nabla)u\cdot\omega\big)(x,t). \tag{14}$$

On $\{r=0\}$ one has $\mathcal{K}=0$. On $\{r>0\}$ one has $\mathcal{K}=r^2(S\xi\cdot\xi)$.

Biot–Savart on $\mathbb{T}^3$ reconstructs u from ω by the periodized Newtonian potential G. Differentiating under the principal value gives S as a Calderón–Zygmund operator of ω. The Constantin–Fefferman representation [3] then yields the pointwise bound, for a dimensional constant c_0 independent of u,

$$|\mathcal{K}(x)| \le c_0 \int_{\mathbb{T}^3} \frac{r(x)^2\, r(y)\, \sin\theta(x,y)}{|x-y|^3}\, \mathrm{d}y, \tag{15}$$

where the integrand is read as 0 if $r(x)=0$ or $r(y)=0$, so that ξ is never invoked on the degeneracy set. (On the torus, $|x-y|$ denotes chordal/periodized distance; the local singularity is the same as in $\mathbb{R}^3$.)

6 High-high versus complementary pairs

Fix $\Omega>0$ as in Theorem A. Split the integral in (15) according to whether $r(y)\ge\Omega$ or not. Write $\mathcal{K}=\mathcal{K}_{\mathrm{hh}}+\mathcal{K}_{\mathrm{c}}$.

High-high. If $r(x)\ge\Omega$ and $r(y)\ge\Omega$, (11) and $\sin\theta\le|\xi(x)-\xi(y)|$ give $\sin\theta/|x-y|^3\le C|x-y|^{-5/2}$. Hence

$$|\mathcal{K}_{\mathrm{hh}}(x)| \le c_0 C\, r(x)^2\, I(x), \qquad I(x) := \int_{\mathbb{T}^3} \frac{r(y)}{|x-y|^{5/2}}\, \mathrm{d}y. \tag{16}$$

The field I is a Riesz potential of order $\beta=1/2$ applied to r. Hardy–Littlewood–Sobolev on $\mathbb{T}^3$ (or $\mathbb{R}^3$) yields

$$\|I\|_{L^3} \le C_{\mathrm{HLS}}\|\omega\|_{L^2}, \tag{17}$$

because $1/3 = 1/2-(1/2)/3$. If $r(x)<\Omega$, the factor $r(x)^2$ already places $\mathcal{K}_{\mathrm{hh}}(x)$ in the complementary class treated next; alternatively one may restrict (16) to $\{r(x)\ge\Omega\}$ and estimate the remainder with $\sin\theta\le 1$ and $r(x)<\Omega$.

Complementary pairs. If $r(y)<\Omega$, use $\sin\theta\le1$ and do *not* form $\xi(y)$. After the usual cancellation in the CZ kernel of S (mean-zero homogeneous kernel of degree -3), the complementary stretching is controlled by the Hardy–Littlewood maximal function of the truncated vorticity:

$$|\mathcal{K}_{\mathrm{c}}(x)| \le c_1 r(x)^2\, M\big(r\,1_{\{r<\Omega\}}\big)(x). \tag{18}$$

In particular, using $r1_{\{r<\Omega\}}\le\Omega$ and the boundedness of $M: L^2\to L^2$,

$$\int_{\mathbb{T}^3} |\mathcal{K}_{\mathrm{c}}|\, \mathrm{d}x \le c_2\Omega\, \|\omega\|_{L^2}\|\omega\|_{L^6}. \tag{19}$$

(The same bound follows from $|S(1_{\{r<\Omega\}}\omega)|\lesssim M(r1_{\{r<\Omega\}})$. No division by r occurs.)

7 Closing the enstrophy inequality

Combining (13), (16), (17) and (19),

$$\int |\mathcal{K}_{\mathrm{hh}}|\,\mathrm{d}x \le c_0 C \int r^2 I\,\mathrm{d}x \le c_0 C\|\omega\|_{L^3}^2\|I\|_{L^3} \le c_3 C\|\omega\|_{L^2}\|\omega\|_{L^6}\|\omega\|_{L^2},$$

where the interpolation $\|\omega\|_{L^3}^2 \le \|\omega\|_{L^2}\|\omega\|_{L^6}$ was used. Thus

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu\|\nabla\omega\|_{L^2}^2 \le c_4(C+\Omega)\,\|\omega\|_{L^2}^2\|\omega\|_{L^6}. \tag{20}$$

Sobolev on $\mathbb{T}^3$ gives $\|\omega\|_{L^6} \le C_{\mathrm{S}}\|\nabla\omega\|_{L^2}$. Young's inequality, for every $\varepsilon > 0$,

$$\|\omega\|_{L^2}^2\|\omega\|_{L^6} \le \frac{\varepsilon}{c_4(C+\Omega)}\|\omega\|_{L^6}^2 + \frac{c_4(C+\Omega)}{4\varepsilon}\|\omega\|_{L^2}^4,$$

so choosing ε small enough to absorb the first term into $\nu\|\nabla\omega\|_{L^2}^2$ produces

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_{L^2}^2 + \nu\|\nabla\omega\|_{L^2}^2 \le K(C,\Omega,\nu)\,\|\omega\|_{L^2}^4, \tag{21}$$

with $K(C,\Omega,\nu)$ the derived coefficient (10). The only data-dependent constants entering K are the two constants Ω and C.

8 Continuation

Energy equality for (1) on $[0,T^*)$ yields

$$\frac{1}{2}\|u(t)\|_{L^2}^2 + \nu\int_0^t \|\nabla u(s)\|_{L^2}^2\,\mathrm{d}s = \frac{1}{2}\|u_0\|_{L^2}^2, \tag{22}$$

hence $\int_0^t \|\omega(s)\|_{L^2}^2\,\mathrm{d}s \le (2\nu)^{-1}\|u_0\|_{L^2}^2$ for every $t < T^*$. Drop the dissipation in (21) and apply Gronwall:

$$\|\omega(t)\|_{L^2}^2 \le \|\omega(0)\|_{L^2}^2 \exp\Big(K\int_0^t \|\omega(s)\|_{L^2}^2\,\mathrm{d}s\Big) \le \|\omega(0)\|_{L^2}^2 \exp\big(K(2\nu)^{-1}\|u_0\|_{L^2}^2\big). \tag{23}$$

The right-hand side is independent of $t < T^*$. Therefore $\sup_{t<T^*}\|\nabla u(t)\|_{L^2} < \infty$. The blow-up alternative (3) forces $T^* = \infty$.

Once the solution exists globally in H^1 with uniformly bounded enstrophy, standard parabolic bootstrapping for (1) (Kato, or L^p-L^q estimates for the Stokes semigroup) upgrades u to $C^\infty(\mathbb{T}^3 \times [0,\infty))$.

This is Theorem A.

9 Construction from the PDE, and the test

The two constants are now built from (1)–(2) and from u_0, then tested against the energy identity. No term is added to the PDE. Degeneracy of ξ on $\{r = 0\}$ is respected.

9.1 Irrotational data

If $\omega_0 \equiv 0$, then u_0 is a harmonic divergence-free field on $\mathbb{T}^3$, hence constant. The solution of (1)–(2) is $u(\,\cdot\,,t) = u_0$, which is smooth for all time. Take $\Omega = 1$ and $C = 0$; the high set is empty. This case is settled and is excluded below.

9.2 The threshold Ω

Fix the intensity threshold independently of the later evolution,

$$\Omega(u_0, \nu) := 1. \tag{24}$$

Any other fixed positive number may replace 1; Beirão da Veiga–Berselli allow an arbitrary $k > 0$. The choice must *not* track $\|\omega(t)\|_\infty$. In particular one cannot take $\Omega(t) > \|\omega(t)\|_\infty$, which would empty the high set and make (11) vacuous by circular use of a bound on ω. The value (24) depends on (u_0, ν) only in that $\nu > 0$ and u_0 is given; it is a universal positive threshold.

9.3 The modulus $C(t)$ along the classical solution

Let $E(t) := \{x \in \mathbb{T}^3 : r(x,t) \geq \Omega\}$. On each compact subinterval of $[0, T^*)$ the solution is smooth, so $E(t)$ is closed and $\xi(t)$ is smooth on a neighbourhood of $E(t)$ whenever $E(t)$ is nonempty (because $r \geq 1$ there). Define the running $\frac{1}{2}$-Hölder modulus on the high set by

$$C(t) := \begin{cases} 0 & \text{if } E(t) = \emptyset, \\ \sup\left\{ \frac{|\xi(x,t) - \xi(y,t)|}{|x-y|^{1/2}} : x, y \in E(t),\ x \neq y \right\} & \text{if } E(t) \neq \emptyset. \end{cases} \tag{25}$$

Local smoothness gives $C(t) < \infty$ for every $t < T^*$. The constant required by Theorem A is the *uniform* value

$$C(u_0, \nu) := \sup_{t \in [0, T^*)} C(t) \in [0, \infty]. \tag{26}$$

This is constructed from the PDE in the only non-circular sense available: it is the actual modulus of the classical solution on its maximal interval. Theorem A applies if and only if this supremum is finite.

9.4 Direction equation on $\{r > 0\}$

On $\{r > 0\}$, $|\xi| = 1$ and $(\omega \cdot \nabla)u = S\omega$. The chain rule in (4) yields

$$D_t \xi = P_\xi(S\xi) + \nu\, r^{-1} P_\xi(\Delta\omega), \tag{27}$$

where $D_t = \partial_t + u \cdot \nabla$ and $P_\xi v := v - (\xi \cdot v)\xi$. Expanding $\Delta\omega = \Delta(r\xi)$ gives the equivalent viscous form

$$r^{-1} P_\xi(\Delta\omega) = \Delta\xi + 2(\nabla \log r) \cdot \nabla\xi, \tag{28}$$

in which $\xi \cdot \Delta\xi = -|\nabla\xi|^2$. The logarithmic gradient is used only on $\{r \geq \Omega\}$, where $|\nabla \log r| \leq \Omega^{-1}|\nabla\omega|$. Thus on $E(t)$

$$|\nabla\xi| \leq \Omega^{-1}|\nabla\omega|. \tag{29}$$

No division by r occurs on $\{r = 0\}$.

9.5 Morrey reduction

Morrey's embedding on $\mathbb{T}^3$ gives $W^{1,6} \hookrightarrow C^{0,1/2}$. Let $\chi \in C^\infty(\mathbb{R})$ satisfy $\chi(s) = 0$ for $s \leq 1$ and $\chi(s) = 1$ for $s \geq 2$, and set $v(x,t) := \chi\big(r(x,t)/\Omega\big)\,\xi(x,t)$ on $\{r > 0\}$, with $v = 0$ on $\{r = 0\}$. Then v is a globally defined smooth field on $[0, T^*)$, $v = \xi$ on $\{r \geq 2\Omega\}$, and

$$|\nabla v| \leq C_\chi\, \Omega^{-1}|\nabla\omega|. \tag{30}$$

Hence a finite bound

$$\sup_{t < T^*} \|\nabla\omega(t)\|_{L^6(\mathbb{T}^3)} < \infty \tag{31}$$

would imply $\sup_{t<T^*} C(t) < \infty$ after replacing Ω by 2Ω in Theorem A. Lipschitz control of ξ would require $\nabla\omega \in L^\infty_t L^\infty_x$, which is Beale–Kato–Majda and is strictly stronger than (31).

9.6 What the energy identity supplies

Energy equality (22) yields only

$$u \in L^\infty(0,T^*;L^2) \cap L^2(0,T^*;H^1), \qquad \omega \in L^2(0,T^*;L^2).$$

It does *not* yield $\nabla\omega \in L^\infty_t L^6$, nor even $\nabla\omega \in L^2_t L^2$. The latter would be an enstrophy bound, which is what Theorem A is designed to produce after C is known to be finite. Using (31) as a hypothesis to bound C is therefore circular at the energy level.

A differential inequality for $C(t)$ from (27) contains the stretching $P_\xi(S\xi)$ and the viscous term $\nu\Omega^{-1}P_\xi(\Delta\omega)$ on $E(t)$. The strain S is a Calderón–Zygmund image of ω. Neither term is controlled in $L^\infty_t L^6$ by $\|\omega\|_{L^2_t L^2_x}$ alone. The same obstruction persists if $\Omega = 1$ is replaced by any other fixed positive threshold depending only on $\|u_0\|_{H^s}$ and ν.

9.7 Result of the test

Theorem B (construction from the PDE does not close)**.** *Let* $u_0 \in C^\infty(\mathbb{T}^3;\mathbb{R}^3)$ *be divergence-free and* $\nu > 0$*, and let* u *be the classical solution of* (1)–(2) *on* $[0,T^*)$*. Define* Ω *by* (24) *and* $C(u_0,\nu)$ *by* (26)*. Then:*

1. *If* $\omega_0 \equiv 0$*, then* $T^* = \infty$ *and* u *is globally smooth.*
2. *If* $\omega_0 \not\equiv 0$*, then* $C(t) < \infty$ *for every* $t < T^*$*, but the energy identity for* (1) *does not imply* $C(u_0,\nu) < \infty$*. In particular it does not imply* (31)*.*
3. *Consequently Theorem A cannot be applied from the energy class alone. The test does not prove* $T^* = \infty$*, and it does not produce a finite-time singularity.*

Local smoothness on compact subintervals of $[0,T^*)$ always gives a finite modulus on each such interval. Uniformity up to T^* is the whole problem. Values $\beta < 1/2$ are not known to suffice [2].

No term was added to (1). Degeneracy of ξ on $\{r = 0\}$ was respected throughout. The Clay problem, as stated by Fefferman [4], remains open.

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